



A combined experimental and numerical investigation of flow dynamic in a methane reformer filled with α -Al₂O₃-supported catalyst



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ABSTRACT

In this study, the combination of numerical simulations and experiments have been carried out to investigate a flow dynamics in a methane reformer that filled with the porous nickel-based catalyst. CFD-modeling was performed in commercial software ANSYS Fluent. The high density meshes were generated for numerical investigation. The numerical simulation is performed for the wide range of the Reynolds numbers. The experimental investigation allowed obtaining an array of experimental data for verification of a numerical model. The comparison of the pressure drops between experimental and simulation results show a good correlation with divergence of results less than 7–8%. To determine the effect of porosity properties of the medium on numerical results, two cases of CFD modeling were realized (with taking into account the porous medium properties and without it). The discrepancy between results is increasing with an increase of the gas flow rate, while for the low flow rates the results are almost similar. The local resistance coefficient of a packed-bed as a function of inlet velocity were obtained. It was found that the local resistance coefficient in the velocity range from 0.5 m/s to 3 m/s varies more than twice.

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1. Introduction

About 90% of hydrogen in the world is produced from fossil fuel sources. Hydrogen production plants have the advantage of using flexible feedstock, such as fossil fuels [1–3], biomass, or water. Many commercial technologies for hydrogen production have been developed in industry, e.g., steam reforming [4,5], partial oxidation, biomass gasification, and alkaline electrolyzer. Specifically, steam methane reforming (SMR) is one of the most common hydrogen production technologies from natural gas owing to the highest thermal efficiency in the range 60–85%, based on the higher heating value [6,7]. The main elements in the steam methane reforming process is a methane reformers that filled with various catalysts. In addition, in recent years, a wide interest among engineers and scientists in the different countries has caused a way to increase the energy efficiency of the fuel-consuming equipments as a thermo-chemical waste-heat recuperation by steam methane reforming [8–10].

The chemical reactions of overall steam methane reforming (SMR) are a well-studied, both theoretically [11–15] and experimentally [16–19]. It is known that different types of the catalysts are applied to improve the efficiency of this chemical process. The nickel (Ni) based catalysts are widely used in large-scale

hydrogen production [15,16,20,21]. Moreover, the noble metals such as ruthenium (Ru) [22], rhodium (Rh) [23], palladium (Pd) [24], iridium (Ir) [11] and platinum (Pt) [25] are also used as an active element in the catalysts for steam methane reforming.

Many scientific works are devoted to the investigation of the steam methane reforming process. In general, scientific publications are devoted to the design and performance study of the new catalysts [26–29], to the energy and exergic analysis of SMR for the different schemes [2,30,31], to the numerical modeling of SMR [1,32–34]. However, it is known that in addition to the chemical processes, the aerodynamic processes (aerodynamics of the gas flow) play an important role in steam reforming of methane. Due to the extended surface and the porous structure of the catalyst, the fixed bed of the steam methane reformer has the high aerodynamic resistance. The pressure drop is a basic importance for the construction and functioning of the steam methane reformer with the packed bed. The literary review showed that there are the scientific publications, based on the numerical and experimental investigation, for a description of the pressure drop in the reformer with the packed beds [35–38].

The experimental investigations of the pressure drop are performed for various conditions: the shape of the catalyst, the porosity, the filling method, etc. [35,37]. The experimental setup for the pressure drop investigating through packed bed is sufficiently simple and consists mainly of two manometers, a flow meter or a Pitot

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tube, a thermometer, etc. [36]. For the numerical investigations there are two major methods to determine the pressure drop of the reformer packed beds. In the first case, the packed bed is presented as a pseudohomogeneous media, where modified Navier-Stokes equations are used in conjunction with the Ergun pressure drop correlation to account for the fluid-solid interaction [39–41]. In the second one, the packed bed is modeled based on the using of the real geometries of the packed bed and catalyst shape. This approach gives a more detailed description of the gas flow in the packed bed. In the second approach, the Navier-Stokes equations are used to the fluid domain between the catalyst particles [36,42,43].

In recent years, a numerical investigation of the different physical and chemical processes has found wide use among researchers. CFD-modeling allows to use a realistic packing structure for the packed bed and simulates flow dynamics between catalyst particles. The main advantages of CFD modeling of flow dynamics in the packed bed are the low cost of the investigation in comparison with the experiments, as well as an evident visualization of the results. The different commercial and open-source codes are applied for CFD simulation of flow in the packed bed, for example, ANSYS Fluent [44,45], Comsol Multiphysics [46], Autodesk Simulation CFD [47], OpenFOAM (open source).

Dixon and Nijemeisland reported about CFD modeling as a design tool for a fixed-bed reactor [42]. The authors concluded that CFD-modeling of a gas flow through the reformer with the packed bed can provide highly detailed and reliable information about the flow fields. They simulated the gas flow for a moderate simple geometry with low number of the cells in a mesh and the catalyst particles. The exponential growth of the computer power is giving supplementary opportunity of obtaining more detailed and accurate information in the numerical results. In Ref. [42], the catalyst packed bed is filled by the spherical particles. Additionally, most of the studies of flow dynamics in the packed bed consider simple shapes of the catalyst particles: spherical [35,42,43] and cylindrical [35,48].

Bai and co-workers presented the numerical simulation results of the flow fields distributions and the pressure drop in a fixed bed reactor with randomly packed catalyst particles [35]. They reported about the numerical results where the discrete element method (DEM) and computational fluid dynamics (CFD) are coupled to model a fixed bed reactor with low tube-to-particle diameter ratios ($D/d < 4$). The authors considered spherical and cylindrical shapes of catalysts and compared obtained numerical results with experimental data. They showed good correlation between the CFD-modeling and experimental results, but they concluded that with an increasing of gas flow rate in the packed bed the divergence between results is increasing. The packed bed was presented as solid body.

In most of the works mentioned above, the catalyst particles are considered as the solid bodies, but it is known that the catalysts for steam methane reforming is porous medium. Wehinger and co-authors [49] reported that the pore processes have a detectable effect on flow dynamics in the packed bed. They numerically investigated flow dynamics in the porous medium of the catalyst bed via the CFD software STAR-CCM+ for a wide range of Reynolds number from 10 to 2000. Moreover, CFD modeling of gas flow in the porous medium was presented by Gao and Zhu [50] to simulate the flow behaviour and catalytic coupling reaction of carbon monoxide (CO)–diethyl oxalate (DEO). They analyzed a two-dimensional computational domain. Karthik and co-worker [37] set the catalyst bed as a porous medium, but they worked with a simple geometry of a packed bed and number of catalyst particles was very low.

The aim of this study is to develop a computational fluid dynamics (CFD) model of flow dynamics in the fixed packed bed for the real geometry of the packed bed taking into account

porosity of the catalysts particles. For this aim, the meshes with more than 20,000,000 cells are used. To verify the numerical results, the experimental determination of the pressure drop in the reformer is performed for the various shapes of catalyst particles. In this study four shapes of the catalyst particles are considered: a cylinder, a cylinder with internal holes, a Raschig ring, a sphere with internal holes. The experimental and numerical study are performed for the different initial conditions. The commercial ANSYS Fluent CFD code is used for numerical simulation.

With the modern computational tools and high power computing technology, the computational fluid dynamics simulations to predict the pressure drop of the catalyst bed of the steam methane reformer are feasible. Whether they will partly or fully replace experimental investigations will largely depend on the reliability of the turbulence and porous medium models that used, and the comprehensive study of it. Therefore, both numerical and experimental investigation of the flow dynamics process in fixed-bed reformer is a crucial task.

2. Experiments

2.1. Experimental setup

To study fluid dynamics and the pressure drop of the packed bed, the experimental setup shown in Fig. 1 is used. The experimental setup mainly consists of the following elements: 1 - an air heater; 2 - the manometer tubes; 3 - a cylindrical tube with inner acrylic tube; 4 - an instrument board presenting the pressure and flow temperature; 5 - a transparent acrylic tube with catalysts; 6 - a fan/ventilator.

To visualize the packing structure of a catalyst bed, the catalyst particles are placed to transparent acrylic tube. And then catalysts with almost same packing structure are placed to the experimental setup. The gas flow is controlled by a fan rotor speed and measured by a Veterok GRM-12 flow-meter. The electronic manometers are used to measure pressure before and after the packed bed. The measuring positions of the electronic manometers were more than 3 times higher than the tube diameter to ensure stable pressure data.

Each structure of the packed bed is tested under 10–15 different flow rates. For each flow rate, the measurements are repeated at least 3–5 times to quantify experimental precision. The pressure drop is determined as the average of the repetitions. Air is heated by electrical heater up to 600 °C.

2.2. Catalyst shapes and packed bed

The packed beds with structured packing of catalyst particles with the different shapes are presented in Fig. 2. There are four particle shapes: A - a simple cylinder; B - a Raschig ring (cylinder with one hole on axial line); C - a convex cylinder with 7 internal holes; D - a sphere with 7 internal holes.

The catalyst particles are placed into the tube with diameter of 100 mm. The catalysts layer depth is 600 mm. The performances of catalysts are shown in Table 2. For experiment the Ni-Al₂O₃ catalyst is used with the following chemical composition: NiO = 14.5%; SiO₂ = 0.2%; support CaO-MgO-La₂O₃-Al₂O₃. Other properties are presented below: bulk density is 680 kg/m³; surface area is 4.5 m²/g_{cat}; average porosity of catalyst particles is about 41%.

3. CFD modeling

3.1. Geometry and mesh

In the present paper, the family of particle shapes is considered as it is shown in Fig. 2.

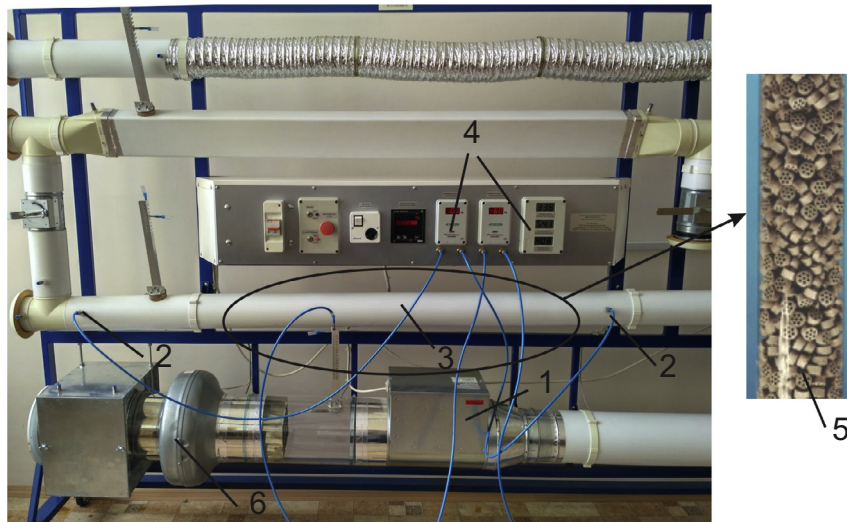


Fig. 1. Photograph of a laboratory experimental setup for investigation of fluid dynamics and pressure drop.

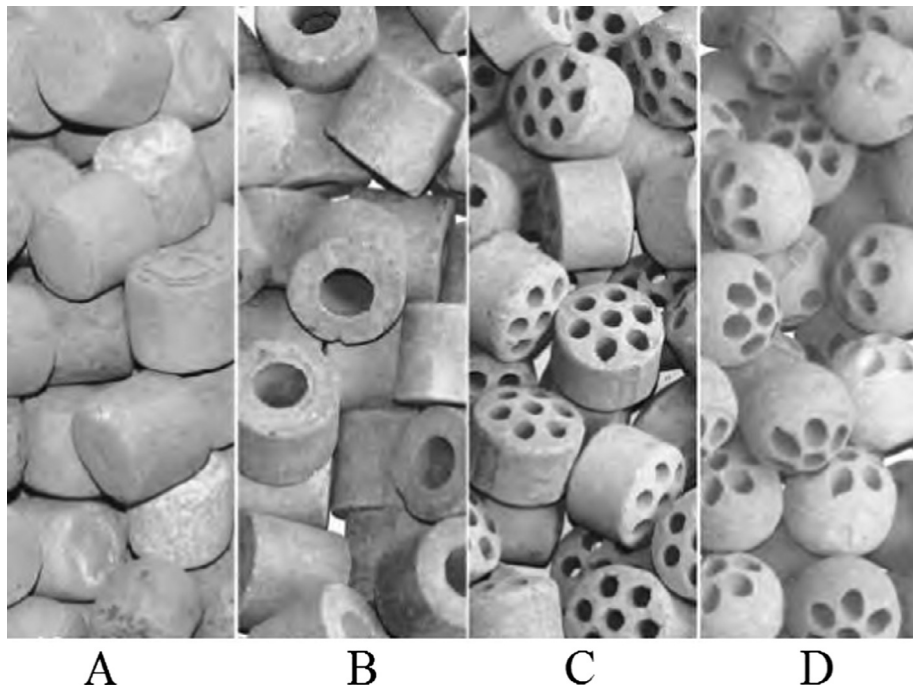


Fig. 2. Photograph of the packed beds filled by the catalysts with different shapes: A – a cylindrical; B – a Raschig ring; C – a convex cylinder with 7 internal holes; D – a sphere with 7 internal holes.

The computational domain has the maximum similarity to the experimental package of catalysts in the reformer. For this numerical investigation, it is assumed that the size of all catalyst particles is the same. In reality, the diameter of the sphere can be different by 0.2–0.4 mm. The computational domain of the packed bed is presented in Fig. 2. The packed bed is 600 mm in length, the diameter is 100 mm. All the cases have same outer diameter and length. To obtain a developed flow for experiments, when a laminar flow takes place, free space is added before the packed layer. The length of this free space is 250 mm.

The mesh structures for the all investigated computational domains presented in Fig. 4. The edge size of the elementary cells is the same for the all computational domains. The meshes for the all cases are generated with the same mesh settings.

To get a more perfect grid, the inflation layers around the catalyst were added in the fluid domain. Moreover, the inflation layer is also added near the tube wall in the fluid domain. The total thickness of the inflation layer is adjusted to get $20 < y^+ < 250$, a requirement for $k-\epsilon$ turbulence model. The mesh independence study is not performed. The maximum number of elementary cells (about 23 millions for our university ANSYS license) are used. Moreover, using the scaled wall function for the modeling is a good solution that can avoid problems with the calculation of the boundary layer for such a complex geometry. ANSYS Fluent has an adapt procedure in the top menu. The mesh adaptation for y^+ was performed by “adapt” procedure in the top menu of ANSYS Fluent. The mesh size is reduced until the y^+ equals $20 < y^+ < 250$.

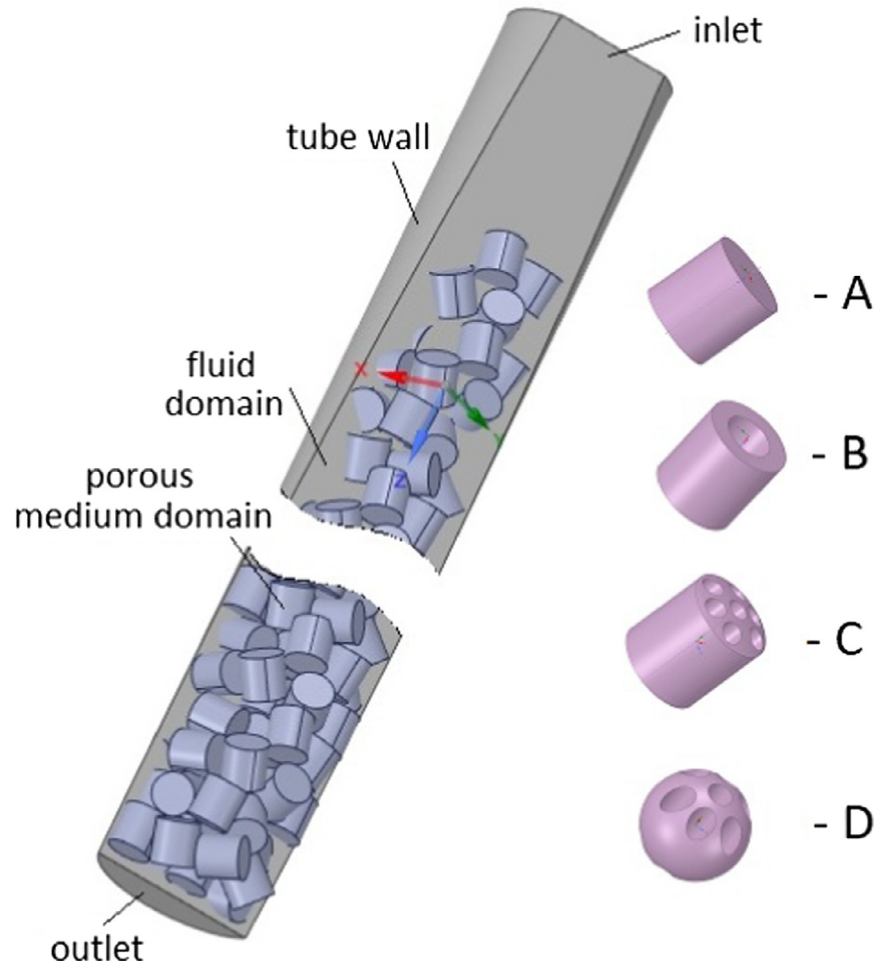


Fig. 3. A computational domain of a packed bed that filled by the catalyst with cylindrical shape (A).

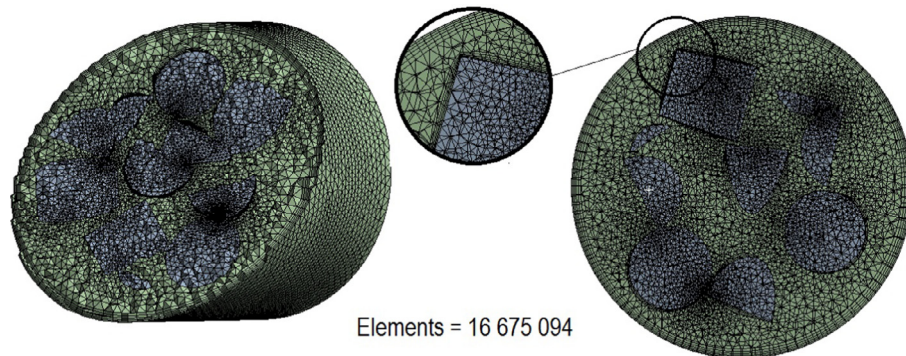


Fig. 4. The mesh structure: (a) catalyst porous media domain; (b) fluid domain.

3.2. Governing equations

Single phase, turbulent, steady flow are modeled by solving the Reynolds-averaged NavierStokes equations. The governing equations are solved in ANSYS Fluent [51,52]. The assumptions are taken for CFD-modeling: the steady-state conditions; no flux of energy due to a mass concentration gradient (no Dufleur effects) [53]; work by viscous forces and by pressure is not done.

The equation for the conservation of mass, or continuity equation, for steady-state conditions can be written as follows:

$$\nabla \cdot (\rho \vec{v}) = 0 \quad (1)$$

For the fluid domain the conservation of momentum is described by the following equations [54]:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\tau}) + \rho \vec{g} + \vec{F} \quad (2)$$

where p is the static pressure; $\rho \vec{g}$ is the gravitational body force.

The stress tensor $\bar{\tau}$ is given by the following equation:

$$\bar{\tau} = \mu \left[(\nabla \vec{v} + \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (3)$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The porous media domain are simulated by the addition of source term for the Eq. (2) that can be expressed as:

$$S_i = - \left(\sum_{j=1}^3 D_{ij} \mu v_j + \sum_{j=1}^3 C_{ij} \frac{1}{2} \rho |v| v_j \right) \quad (4)$$

where S_i is the source term for the i -th (x , y , or z) momentum equation, $|v|$ is the magnitude of the velocity and D and C are prescribed matrices. This momentum sink contributes to the pressure gradient in the porous cell, creating a pressure drop that is proportional to the fluid velocity (or velocity squared) in the cell.

The Eq. (4) can be used for heterogeneous porous medium. However the nickel-based catalysts have a homogeneous porous medium. To recover the case of simple homogeneous porous media the Eq. (4) can be written as:

$$S_i = - \left(\frac{\mu}{\alpha} v_i + C_2 \frac{1}{2} \rho |v| v_i \right) \quad (5)$$

where α is the permeability and C_2 is the inertial resistance factor, simply specify D and C as diagonal matrices with $1/\alpha$ and C_2 , respectively, on the diagonals (and zero for the other elements).

The momentum source term S_i for every dimension (x , y , z) consists of two parts. The first term of the Eq. (4) is a viscous loss term. The second term of the Eq. (4) is an inertial loss term. This momentum sink contributes to the pressure gradient in the porous cell, creating the pressure drop that is proportional to the fluid velocity (or velocity squared) in the cell.

The presented model is solving the standard conservation equations for turbulence quantities in the porous medium. In this default approach, turbulence in the medium is treated as though the solid medium has no effect on the turbulence generation or dissipation rates. This assumption may be reasonable if the medium's permeability is quite large and the geometric scale of the medium does not interact with the scale of the turbulent eddies.

Typically, steam methane reforming is taking place at a wide range of Reynolds numbers. For the numerical model, the RNG k - ϵ model is chosen [54–57]. The Scalable Wall Function is used for RNG k - ϵ model, because there are large numbers of contacting surfaces in the computational domain. The mathematical description of RNG k - ϵ model can be expressed as following:

$$\frac{\partial}{\partial x_i} (\rho k v_i) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right) + G_k + G_b - \rho \epsilon - Y_M + S_k \quad (6)$$

and

$$\begin{aligned} \frac{\partial}{\partial x_i} (\rho \epsilon v_i) = & \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{eff} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \\ & - R_\epsilon + S_\epsilon \end{aligned} \quad (7)$$

In equations for k - ϵ turbulence model, G_k shows the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. The dilatation dissipation term, Y_M , is included in the k Eq. (6) can be calculated according to a proposal by Sarkar [58]:

$$Y_M = 2\rho \epsilon M_t^2 \quad (8)$$

where M_t is the turbulent Mach number.

The quantities α_k and α_ϵ are the inverse effective Prandtl numbers for k and ϵ , respectively. S_k and S_ϵ are user-defined source terms. Usually S_k , S_ϵ and Y_M is used for high-Mach-number flows. In present study the velocity of flow is negligible low. Therefore

these terms can be ignored but the code of ANSYS Fluent is closed and it is impossible change classical equations for k and ϵ .

The main difference between the RNG and standard k - ϵ models lies in the additional term R_ϵ in the ϵ equation. This term can be determined as follows:

$$R_\epsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0) \epsilon^2}{1 + \beta \eta^3} \frac{1}{k} \quad (9)$$

where $\eta \equiv Sk/\epsilon$, $\eta_0 = 4.38$, $\beta = 0.012$.

In computational domains where $\eta < \eta_0$, the R term makes a positive contribution. In regions of large strain rate ($\eta > \eta_0$), however, the R term makes a negative contribution. Therefore, the RNG model is more responsive to the effects of rapid strain and streamline curvature than the standard k - ϵ model, which explains the superior performance of the RNG model for certain classes of flows.

The model constants $C_{1\epsilon}$ and $C_{2\epsilon}$ in k - ϵ turbulence model used by default setup for Fluent solver of ANSYS [59]:

$$C_{1\epsilon} = 1.42; \quad C_{2\epsilon} = 1.68 \quad (10)$$

The performances of the porous medium are determined by the viscous resistance coefficients and the inertial resistance coefficients for each direction. These coefficients, respectively, can be determined as following:

$$\alpha = \frac{D_p^2}{150} \frac{\epsilon^3}{(1 - \epsilon)^2}; \quad (11)$$

$$C_2 = \frac{1.75 \rho (1 - \epsilon)}{D_p} \frac{1}{\epsilon^3}, \quad (12)$$

where D_p – the length of the fixed bed; ϵ – the void fraction. Here the void fraction (ϵ) is taken into account the interparticle spaces and the pellet porosity.

The schematic algorithm of CFD modeling is presented in Fig. 5.

4. Results and discussion

4.1. Verification

The experimental and numerical investigation were performed at same operating conditions. The CFD model was realized via ANSYS Fluent on Xeon E5-2670 Sandy Bridge-EP server with double 20-core 2.6 GHz processors with 32 GB RAM. Convergence for each numerical experiment took approximately 3000 iterations for which the computation time was on the order of 20–23 h. The Reynolds number during both experimental and numerical investigations ranged from about 500 to about 9000. The change in the Reynolds number is done by changing the velocity.

Further, all the experimental data are denoted by the hollow points. The results of numerical simulation are represented by the solid lines. To verify the results obtained with the help of CFD model, they were compared with experimental data for the different packed beds. Fig. 6 shows the comparison of numerical and experimental data for the different packed beds with 600 mm in length. The value of the pressure drop for the numerical results was obtained by Vortex Average function in Fluent solver.

As can be seen from Fig. 6, the differences between the experimental and numerical results are observed for all investigated cases. However, at the velocity range above 0.5 m/s, which corresponds to the real operation parameters of the steam methane reformers, the discrepancies between the results are reduced. The average deviations between the numerical and experimental results in the velocity range above 1 m/s are 6.12% for the cylinder without holes, 2.15% for the Raschig ring, 4.75% for the cylinder with 7 holes, 4.34% for the sphere with 7 holes. Therefore, it can be concluded that the results have a good correlation and the

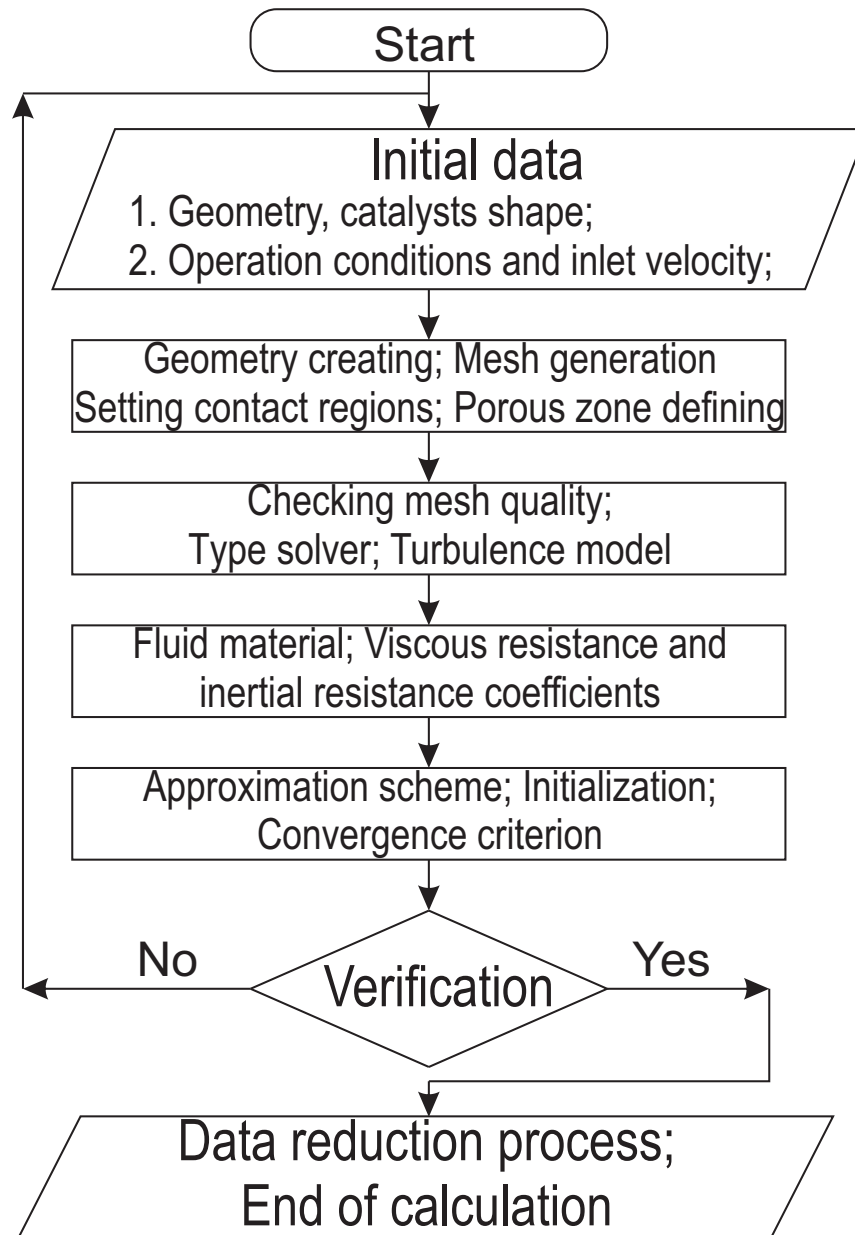


Fig. 5. The algorithm of a fluid flow simulation in the packed bed.

mathematical model is reliable to study the flow dynamics in the packed bed.

4.2. Pressure and velocity contours

Fig. 7 shows the typical CFD simulation results for the packed bed filled with the cylindrical catalysts, including the velocity contours on the catalyst particles (Fig. 7A), the velocity contours into the catalyst particles (Fig. 7B) and the pressure contours on particles (Fig. 7C). Each point on the pressure profile plot corresponds to one spatial position in the packing. Fig. 7 depicts that the flow of gas passes not only between the catalyst particles, but inside them.

The velocity contours in the axial and radial cross-section of the packed bed and are shown in Fig. 8. The velocity magnitude is significantly lower into the particle shapes compared to the velocity magnitude into free space between the cylindrical particles.

4.3. Loss factor

Fig. 9 shows the dependence of the pressure losses per meter of the packed bed for different Reynolds numbers. From Fig. 9 it can be seen that the maximum pressure loss per 1 meter of the packed bed takes place for a cylinder without holes (case A in Table 1), while a minimum pressure loss occurs for a cylinder with 7 holes (case C in Table 1). At the same time, a characteristic quadratic dependence $\Delta P = f(Re^2)$ is observed for all four curves: $\Delta P = f(Re^{1.92})$ for A case; $\Delta P = f(Re^{1.94})$ for B case; $\Delta P = f(Re^{1.93})$ for C case; $\Delta P = f(Re^{1.94})$ for D case.

From the point of view of the classical representation of fluid dynamics, the pressure losses in the catalyst bed can be considered as the local hydraulic losses, which are caused by changes in the shape and size of the channel, which is deforming the flow. As shown by the experimental and numerical results (Figs. 6 and 9),

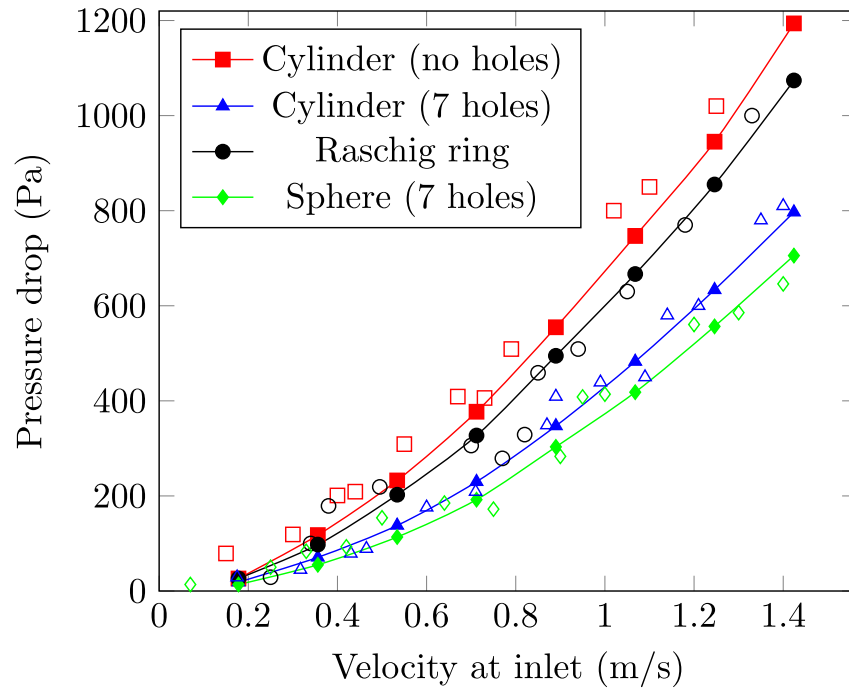


Fig. 6. The comparison of numerical and experimental data: the hollow points – experiment; the solid points and lines – CFD modeling.

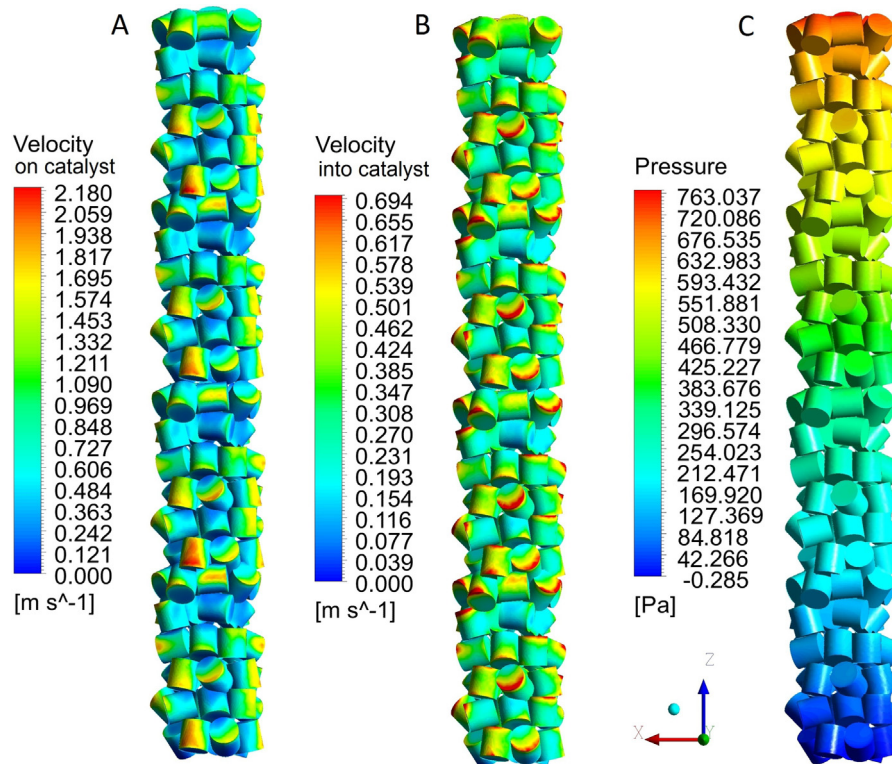


Fig. 7. The velocity and pressure contours.

it can be approximately assumed that the pressure loss through the catalyst bed is proportional to the square of the velocity. For this reason, for the engineering calculations, it is convenient to characterize the fluid resistance of a catalyst bed with a dimensionless quantity K_l , which is called the local resistance coefficient or loss factor.

The local resistance coefficient can be determined experimentally:

$$K_l = \frac{2\Delta P}{\rho v^2}, \quad (13)$$

where ρ – the density of fluid; v – the velocity of fluid.

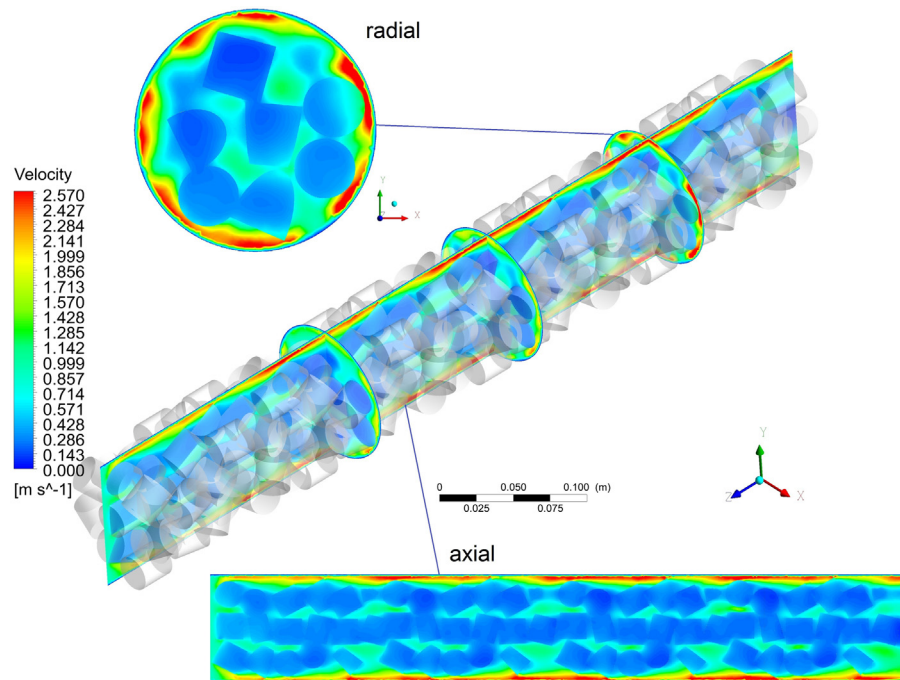


Fig. 8. Simulated velocity magnitude distribution for velocity $u = 0.7$ m/s.

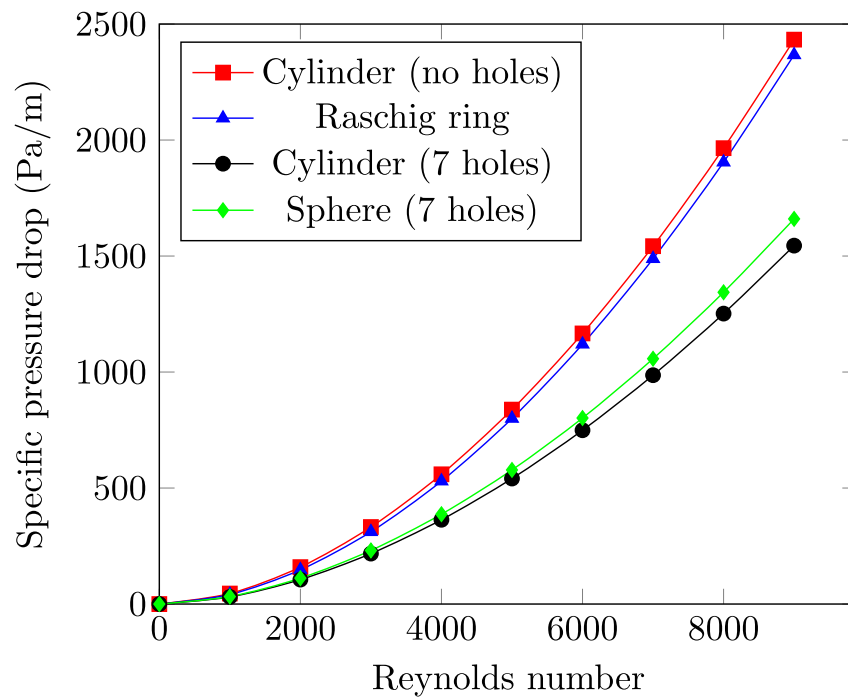


Fig. 9. The variations of the pressure drop versus Reynolds number for the various packed beds.

Table 1
The geometrical characteristics of catalyst.

Catalyst shape	Diameter, mm	Thickness, mm	Holes, number/diameter
A	25	25	–
B	25	25	1×13
C	25	25	7×5
D	25	–	7×5

To determine the local resistance coefficient, the effect of length of the packed bed on the pressure drop is investigated. For this purpose, the different lengths of the packed bed are considered. The computational domains (Fig. 3) were changed, but the mesh structures were same as in previous cases. Fig. 10 shows the variations of the pressure drop as a function of the packed beds length for $Re = 6000$.

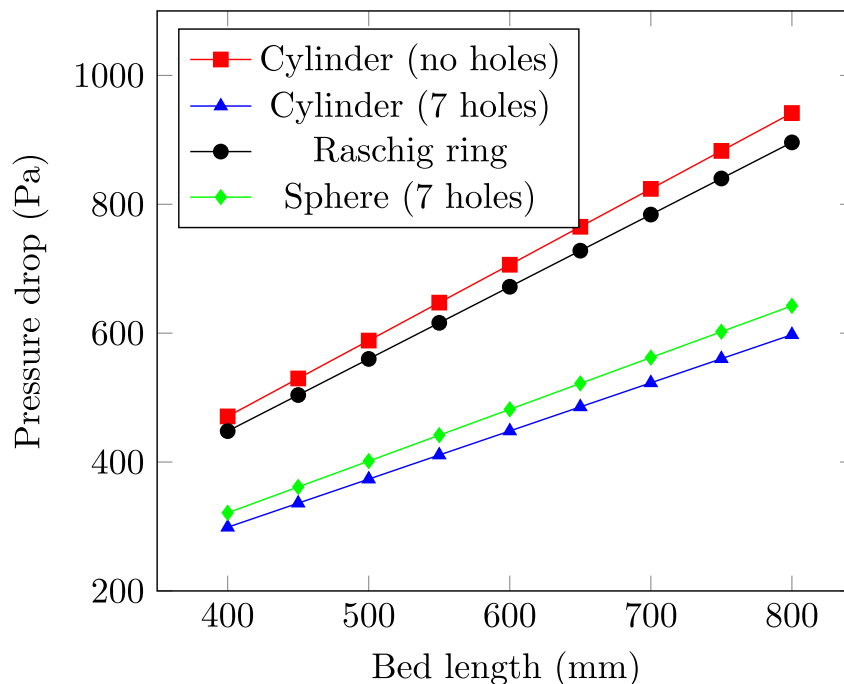


Fig. 10. The variations of the pressure drop for the different packed beds length for $Re = 6000$.

As it can be seen from Fig. 10, the dependence of the pressure loss for the different lengths of the catalyst bed is approximately linear. This fact allows to conclude that the value of the local resistance coefficient for the catalyst bed is determined by the amount of catalyst in the packed bed and the shape of catalyst particle.

Based on the results which are presented in Fig. 10, and also on the basis of the expression (13), the values of the local resistance coefficients were determined for various cases (Table 1). The variations of the local resistance coefficient for 1 meter of the packed

bed versus the average velocity for the different packed beds are shown in Fig. 11.

As seen in Fig. 11, the loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s. Usually, when flowing around solid non-porous bodies, the coefficient K_l depends weakly on the velocity. This effect of velocity on the coefficient K_l can be explained by the fact that with increasing velocity, the amount of gas that passes through the pores in catalyst increases. At the same time, when the gas velocity is low, the fluid

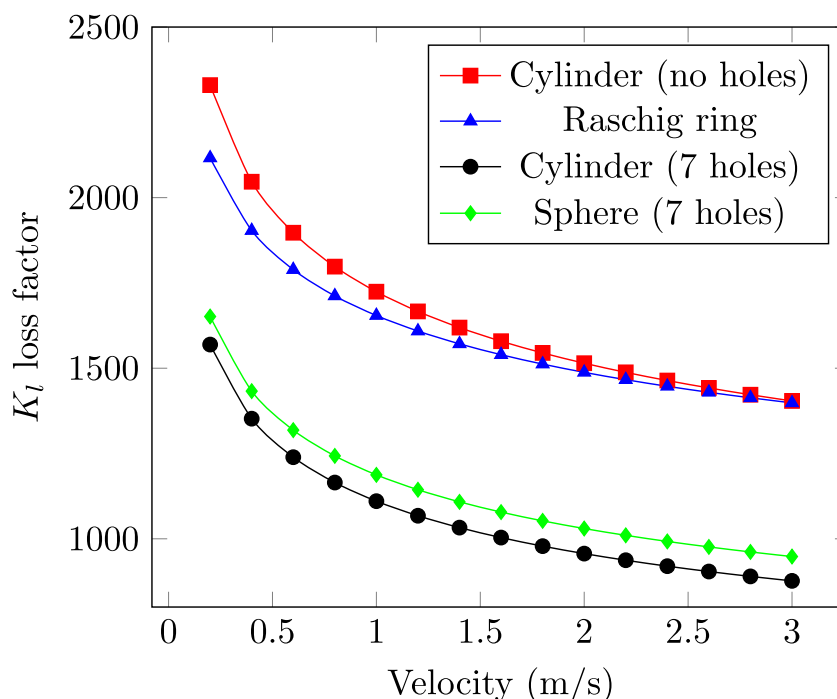


Fig. 11. The variations of the local resistance coefficients versus the velocity for the different packed beds for $Re = 6000$.

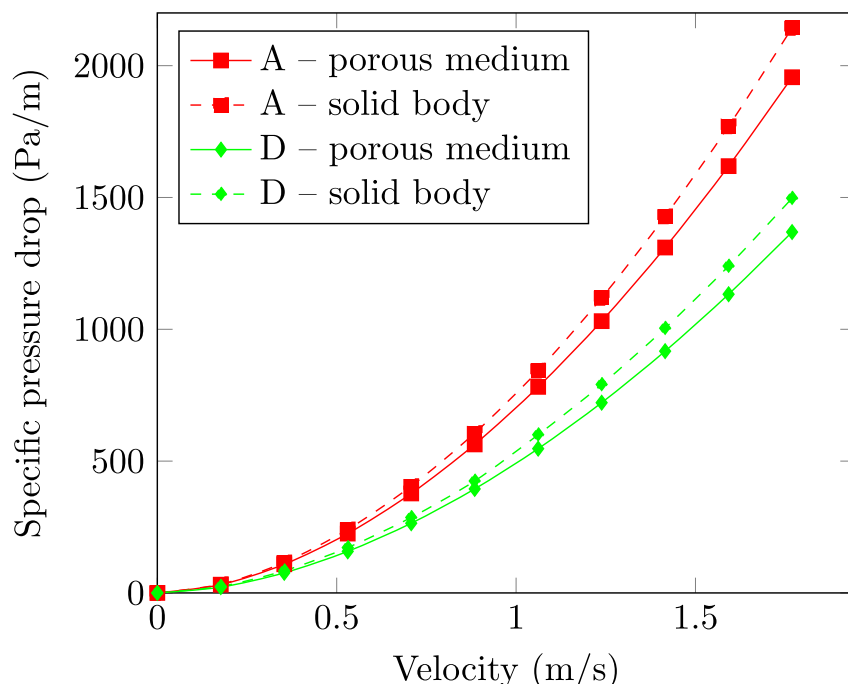


Fig. 12. The variations of the specific pressure drop versus the velocity with taking into account of the porous medium properties and without it.

energy is not enough to percolate through the porous structure. The next paragraph presents the results of investigation the velocity distribution in porous media.

4.4. Porous media

There are scientific articles dealing with the study of flow dynamics in a packed bed that treat catalyst particles as a solid body [35]. In this paragraph, the effect of taking into account of the porous medium properties on the numerical results is reported. For this purpose, the computational domains that presented above were considered. The two type of the packed beds are considered: A – cylinder without holes; D – sphere with 7 holes.

Fig. 12 show the discrepancy between the results obtained with taking into account of the porous medium properties (solid line) and without it (dashed line). As it can be seen from Fig. 12, the discrepancy between the results is increasing with an increase of the velocity magnitude, while for the low velocity the results are almost similar. Therefore, for saving calculation time at low velocity, the packed bed can be considered as a solid body and the porous media properties can be ignored in the numerical model. However, for the velocity above 1 m/s, the porous medium properties must be taken into account, because the discrepancies are more than 10%.

5. Conclusion

The numerical simulation and experimental investigation of fluid dynamics in the packed bed of Ni-Al₂O₃ catalysts with the different shapes are performed in this paper. The CFD model is realized via ANSYS Fluent, Academic Research version which allows to use large numbers of catalysts particles without any limitation. Both numerical and experimental study were performed for the large L/d ratio of the packed bed for a wide range of Reynolds number. The algorithm of the fluid flow simulation with taking into account of the porous medium properties of catalyst particles is depicted. Four types of the catalysts shapes are considered. The good correlation between the numerical and experimental results

is achieved for all experiments with an average error of less than 8%.

A characteristic quadratic dependence $\Delta P = f(v^2)$ is observed for all catalyst types. The effect of the packed bed depth on the pressure drop is investigated. The dependence of the pressure loss for the different depths of the catalyst bed is approximately linear. This fact allowed to conclude that the local resistance coefficient for the catalyst bed is constant for all investigated L/d ratios. The loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s.

To determine the effect of porosity properties of the medium on numerical results, two cases of CFD modeling were realized. First, with taking into account the porous medium properties; second, without it. The discrepancy between the results is increasing with an increase of the velocity, while for the low velocity the results are almost similar. As a recommendation for the engineering solutions, it can be concluded that for saving calculation time at the low velocity, the packed bed can be considered as a solid body and the porous media properties can be ignored in the numerical model. However, for the velocity above 1 m/s, the porous medium properties must be taken into account, because the discrepancies are more than 10%.

Conflict of interest

The author declared that there is no conflict of interest.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ijheatmasstransfer.2018.12.150>.

References

- [1] A. Tran, A. Aguirre, H. Durand, M. Crose, P.D. Christofides, CFD modeling of a industrial-scale steam methane reforming furnace, *Chem. Eng. Sci.* 171 (2017) 576–598.

- [2] M.S. Nobandegani, M.R.S. Birjandi, T. Darbandi, M.M. Khalilipour, F. Shahraki, D. Mohebbi-Kalhor, An industrial steam methane reformer optimization using response surface methodology, *J. Nat. Gas Sci. Eng.* 36 (2016) 540–549.
- [3] G. Diglio, D.P. Hanak, P. Bareschino, E. Mancusi, F. Pepe, F. Montagnaro, V. Manovic, Techno-economic analysis of sorption-enhanced steam methane reforming in a fixed bed reactor network integrated with fuel cell, *J. Power Sour.* 364 (2017) 41–51.
- [4] G. Ji, M. Zhao, G. Wang, Computational fluid dynamic simulation of a sorption-enhanced palladium membrane reactor for enhancing hydrogen production from methane steam reforming, *Energy* 147 (2018) 884–895.
- [5] C.-H. Kim, J.-Y. Han, S. Kim, B. Lee, H. Lim, K.-Y. Lee, S.-K. Ryi, Hydrogen production by steam methane reforming in a membrane reactor equipped with a Pd composite membrane deposited on a porous stainless steel, *Int. J. Hydrogen Energy* 43 (15) (2018) 7684–7692.
- [6] D. Czyłkowski, B. Hrycak, M. Jasiński, M. Dors, J. Mizeraczyk, Microwave plasma-based method of hydrogen production via combined steam reforming of methane, *Energy* 113 (2016) 653–661.
- [7] K.S. Santhanam, R.J. Press, M.J. Miri, A.V. Bailey, G.A. Takacs, Introduction to Hydrogen Technology, John Wiley & Sons, 2017.
- [8] D. Pashchenko, First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming, *Energy* 143 (2018) 478–487.
- [9] D. Pashchenko, Energy optimization analysis of a thermochemical exhaust gas recuperation system of a gas turbine unit, *Energy Convers. Manage.* 171 (2018) 917–924.
- [10] D. Pashchenko, Thermochemical recovery of heat contained in flue gases by means of bioethanol conversion, *Therm. Eng.* 60 (6) (2013) 438–443.
- [11] F. Watanabe, I. Kaburaki, N. Shimoda, S. Satokawa, Influence of nitrogen impurity for steam methane reforming over noble metal catalysts, *Fuel Process. Technol.* 152 (2016) 15–21.
- [12] A. Kumar, M. Baldea, T.F. Edgar, A physics-based model for industrial steam-methane reformer optimization with non-uniform temperature field, *Comput. Chem. Eng.* 105 (2017) 224–236.
- [13] C. Herce, C. Cortés, S. Stendardo, Computationally efficient CFD model for scale-up of bubbling fluidized bed reactors applied to sorption-enhanced steam methane reforming, *Fuel Process. Technol.* 167 (2017) 747–761.
- [14] S. Abbas, V. Dupont, T. Mahmud, Modelling of H₂ production in a packed bed reactor via sorption enhanced steam methane reforming process, *Int. J. Hydrogen Energy* 42 (30) (2017) 18910–18921.
- [15] S. Abbas, V. Dupont, T. Mahmud, Kinetics study and modelling of steam methane reforming process over a NiO/Al₂O₃ catalyst in an adiabatic packed bed reactor, *Int. J. Hydrogen Energy* 42 (5) (2017) 2889–2903.
- [16] J. Xu, G.F. Froment, Methane steam reforming, methanation and water-gas shift: I. Intrinsic kinetics, *AIChE J.* 35 (1) (1989) 88–96.
- [17] K. Hou, R. Hughes, The kinetics of methane steam reforming over a Ni/ α -Al₂O₃ catalyst, *Chem. Eng. J.* 82 (1) (2001) 311–328.
- [18] H. Wang, D.W. Blaylock, A.H. Dam, S.E. Liland, K.R. Rout, Y.-A. Zhu, W.H. Green, A. Holmen, D. Chen, Steam methane reforming on a Ni-based bimetallic catalyst: density functional theory and experimental studies of the catalytic consequence of surface alloying of Ni with Ag, *Catal. Sci. Technol.* 7 (8) (2017) 1713–1725.
- [19] Y. Yoon, H. Kim, J. Lee, Enhanced catalytic behavior of Ni alloys in steam methane reforming, *J. Power Sour.* 359 (2017) 450–457.
- [20] A. Di Giuliano, J. Gurr, R. Massacesi, K. Gallucci, C. Courson, Sorption enhanced steam methane reforming by Ni–CaO materials supported on mayenite, *Int. J. Hydrogen Energy* 42 (19) (2017) 13661–13680.
- [21] H.W.A. El Hava, S.-T.B. Lundin, N.S. Patki, J.D. Way, Steam methane reforming in a Pd–Au membrane reactor: long-term assessment, *Int. J. Hydrogen Energy* 41 (24) (2016) 10193–10201.
- [22] E.H. Cho, K.Y. Koo, H.W. Lee, Y.-K. Park, W.L. Yoon, C.H. Ko, Preparation of egg-shell-type Ni/Ru bimetal alumina pellet catalysts: steam methane reforming for hydrogen production, *Int. J. Hydrogen Energy* 42 (29) (2017) 18350–18357.
- [23] R. Chakrabarti, J.L. Colby, L.D. Schmidt, Effects of biomass inorganics on rhodium catalysts: I. Steam methane reforming, *Appl. Catal. B: Environ.* 107 (1) (2011) 88–94.
- [24] C.-H. Kim, J.-Y. Han, H. Lim, K.-Y. Lee, S.-K. Ryi, Methane steam reforming using a membrane reactor equipped with a Pd-based composite membrane for effective hydrogen production, *Int. J. Hydrogen Energy* 43 (11) (2018) 5863–5872.
- [25] V.K. Jaiswar, S. Katheria, G. Deo, D. Kunzru, Effect of Pt doping on activity and stability of Ni/MgAl₂O₄ catalyst for steam reforming of methane at ambient and high pressure condition, *Int. J. Hydrogen Energy* 42 (30) (2017) 18968–18976.
- [26] S. Itkulova, Y. Nurmakanov, S. Kussanova, Y. Boleubayev, Production of a hydrogen-enriched syngas by combined CO₂-steam reforming of methane over Co-based catalysts supported on alumina modified with zirconia, *Catal. Today* 299 (2018) 272–279.
- [27] Y. Hiramitsu, M. Demura, Y. Xu, M. Yoshida, T. Hirano, Catalytic properties of pure Ni honeycomb catalysts for methane steam reforming, *Appl. Catal. A: Gen.* 507 (2015) 162–168.
- [28] Z. Boukha, M. Gil-Calvo, B. de Rivas, J.R. González-Velasco, J.I. Gutiérrez-Ortiz, R. López-Fonseca, Behaviour of Rh supported on hydroxyapatite catalysts in partial oxidation and steam reforming of methane: on the role of the speciation of the Rh particles, *Appl. Catal. A: Gen.* 556 (2018) 191–203.
- [29] S. Akbari-Emadabadi, M. Rahimpour, A. Hafizi, P. Keshavarz, Production of hydrogen-rich syngas using Zr modified Ca–Co bifunctional catalyst-sorbent in chemical looping steam methane reforming, *Appl. Energy* 206 (2017) 51–62.
- [30] A. Tran, M. Pont, M. Crose, P.D. Christofides, Real-time furnace balancing of steam methane reforming furnaces, *Chem. Eng. Res. Des.* 134 (2018) 238–256.
- [31] L. Zhu, M. Zhou, C. Shao, J. He, Comparative exergy analysis between liquid fuels production through carbon dioxide reforming and conventional steam reforming, *J. Clean. Prod.* 192 (2018) 88–98.
- [32] A.G. Dixon, Local transport and reaction rates in a fixed bed reactor tube: endothermic steam methane reforming, *Chem. Eng. Sci.* 168 (2017) 156–177.
- [33] D. Pashchenko, Effect of the geometric dimensionality of computational domain on the results of CFD-modeling of steam methane reforming, *Int. J. Hydrogen Energy* 43(18), 8662–8673.
- [34] L. Lao, A. Aguirre, A. Tran, Z. Wu, H. Durand, P.D. Christofides, Cfd modeling and control of a steam methane reforming reactor, *Chem. Eng. Sci.* 148 (2016) 78–92.
- [35] H. Bai, J. Theuerkauf, P.A. Gillis, P.M. Witt, A coupled DEM and CFD simulation of flow field and pressure drop in fixed bed reactor with randomly packed catalyst particles, *Ind. Eng. Chem. Res.* 48 (8) (2009) 4060–4074.
- [36] H. Calis, J. Nijenhuis, B. Paikert, F. Dautzenberg, C. Van Den Bleek, CFD modelling and experimental validation of pressure drop and flow profile in a novel structured catalytic reactor packing, *Chem. Eng. Sci.* 56 (4) (2001) 1713–1720.
- [37] G. Karthik, V.V. Buwa, Effect of particle shape on fluid flow and heat transfer for methane steam reforming reactions in a packed bed, *AIChE J.* 63 (1) (2017) 366–377.
- [38] T. Maffei, G. Gentile, S. Rebughini, M. Bracconi, F. Manelli, S. Lipp, A. Cuoci, M. Maestri, A multiregion operator-splitting CFD approach for coupling microkinetic modeling with internal porous transport in heterogeneous catalytic reactors, *Chem. Eng. J.* 283 (2016) 1392–1404.
- [39] J. Papageorgiou, G. Froment, Simulation models accounting for radial voidage profiles in fixed-bed reactors, *Chem. Eng. Sci.* 50 (19) (1995) 3043–3056.
- [40] O. Bey, G. Eigenberger, Fluid flow through catalyst filled tubes, *Chem. Eng. Sci.* 52 (8) (1997) 1365–1376.
- [41] C.O. Castillo-Araiza, F. Lopez-Isunza, Hydrodynamic models for packed beds with low tube-to-particle diameter ratio, *Int. J. Chem. React. Eng.* 6(1).
- [42] A.G. Dixon, M. Nijemeisland, CFD as a design tool for fixed-bed reactors, *Ind. Eng. Chem. Res.* 40 (23) (2001) 5246–5254.
- [43] M. Klöcker, E.Y. Kenig, R. Piechota, S. Burghoff, Y. Egorov, CFD-based study on hydrodynamics and mass transfer in fixed catalyst beds, *Chem. Eng. Technol.* 28 (1) (2005) 31–36.
- [44] A. Tran, A. Aguirre, M. Crose, H. Durand, P.D. Christofides, Temperature balancing in steam methane reforming furnace via an integrated CFD/data-based optimization approach, *Comput. Chem. Eng.* 104 (2017) 185–200.
- [45] J. Chen, L. Yan, W. Song, D. Xu, Methane steam reforming thermally coupled with catalytic combustion in catalytic microreactors for hydrogen production, *Int. J. Hydrogen Energy* 42 (1) (2017) 664–680.
- [46] O. Corigliano, P. Fragiocomo, Numerical simulations for testing performances of an indirect internal CO₂ reforming solid oxide fuel cell system fed by biogas, *Fuel* 196 (2017) 378–390.
- [47] M. Unsel, C. Szepanski, A. Lindermeir, W. Maus-Friedrichs, S. Dahle, Desulfurization of biogas via dielectric barrier discharges, *Chem. Eng. Technol.* 40 (2) (2017) 333–339.
- [48] W. Zhang, Experimental and computational analysis of random cylinder packings with applications.
- [49] G.D. Wehinger, F. Klippel, M. Kraume, Modeling pore processes for particle-resolved CFD simulations of catalytic fixed-bed reactors, *Comput. Chem. Eng.* 101 (2017) 11–22.
- [50] X. Gao, Y.-P. Zhu, Z.-h. Luo, CFD modeling of gas flow in porous medium and catalytic coupling reaction from carbon monoxide to diethyl oxalate in fixed-bed reactors, *Chem. Eng. Sci.* 66 (23) (2011) 6028–6038.
- [51] D. Pashchenko, Comparative analysis of hydrogen/air combustion cfd-modeling for 3d and 2d computational domain of micro-cylindrical combustor, *International Journal of Hydrogen Energy* 42 (49) (2017) 29545–29556.
- [52] D. Pashchenko, Numerical study of steam methane reforming over a pre-heated Ni-based catalyst with detailed fluid dynamics, *Fuel* 236 (2019) 686–694.
- [53] M. Lücke et al., Influence of the Dufour effect on convection in binary gas mixtures, *Phys. Rev. E* 52 (1) (1995) 642.
- [54] G.K. Batchelor, An Introduction to Fluid Dynamics, Cambridge University Press, 2000.
- [55] H. Yilmaz, O. Cam, S. Tanoz, I. Yilmaz, Effect of different turbulence models on combustion and emission characteristics of hydrogen/air flames, *International Journal of Hydrogen Energy* 42 (40) (2017) 25744–25755.
- [56] A. Mrakin, A. Selivanov, P. Batrakov, D. Sotnikov, The chemical energy unit partial oxidation reactor operation simulation modeling, *Journal of Physics: Conference Series*, vol. 944, IOP Publishing, 2018, p. 012082.
- [57] D. Pashchenko, ANSYS fluent CFD modeling of solar air-heater thermoaerodynamics, *Appl. Solar Energy* 54 (1) (2018) 32–39.
- [58] S. Sarkar, L. Balakrishnan, Application of a Reynolds stress turbulence model to the compressible shear layer, in: 21st Fluid Dynamics, Plasma Dynamics and Lasers Conference, 1990, p. 1465.
- [59] V. Yakhot, S.A. Orszag, Renormalization group analysis of turbulence. I. Basic theory, *J. Sci. Comput.* 1 (1) (1986) 3–51.